

Initiating Search

November 10, 2025, 2:55 PM

• Search:

CAS Lexicon Search:

Wacker oxidation - Preferred Concept

Search Tasks

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Yield:	90-100%, 80-89%, 70-79%	
Document Type:	Journal	

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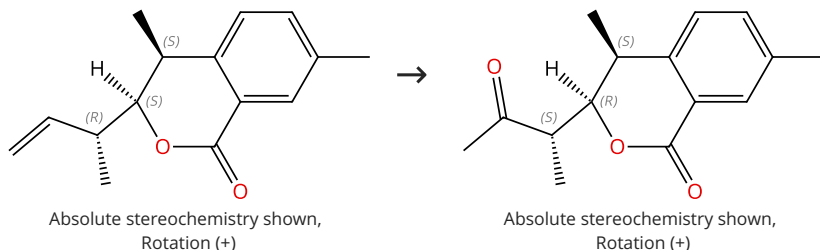
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Reactions (50)

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Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%



31-614-CAS-34288751

Steps: 1 Yield: 100%

Divergent Asymmetric Synthesis of Panowamycins, TM-135, and Veramycin F Using C-H Insertion with Donor/Donor Carbenes

By: Bergstrom, Benjamin D.; et al

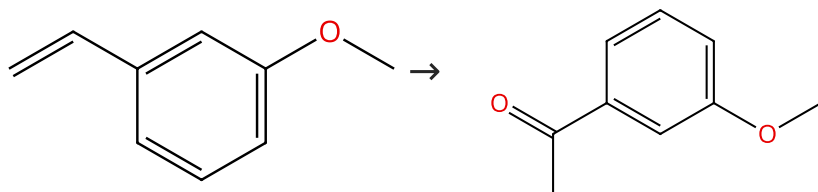
Angewandte Chemie, International Edition (2022), 61(25), e202203072.

1.1 **Reagents:** Oxygen
Catalysts: Palladium chloride, Cuprous chloride
Solvents: Dimethylformamide, Water; 2 d, rt

Experimental Protocols

Scheme 2 (1 Reaction)

Steps: 1 Yield: 100%


 Suppliers (52)

 Suppliers (90)

31-614-CAS-24718987

Steps: 1 Yield: 100%

Aluminum Hydroxide Secondary Building Units in a Metal-Organic Framework Support Earth-Abundant Metal Catalysts for Broad-Scope Organic Transformations

By: Feng, Xuanyu; et al

ACS Catalysis (2019), 9(4), 3327-3337.

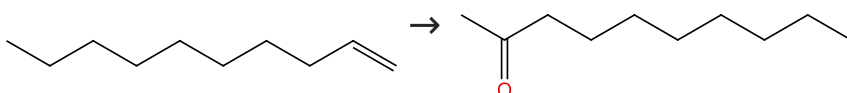
1.1 **Reagents:** 1,1,1,3,5,5,5-Heptamethyltrisiloxane
Catalysts: 2725020-49-3
Solvents: Ethanol; 12 h, 60 °C

1.2 **Reagents:** Potassium fluoride; 1 h, rt

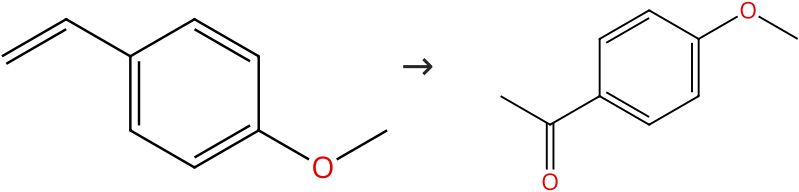
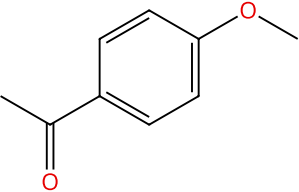
Experimental Protocols

Scheme 3 (2 Reactions)

Steps: 1 Yield: 98-100%

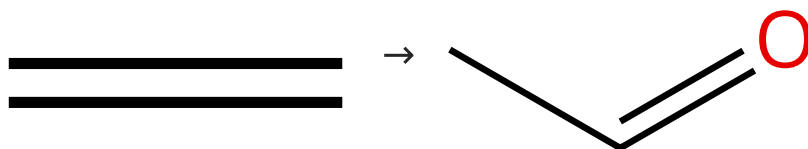

 Suppliers (76)

 Suppliers (67)

31-502-CAS-12617335 Steps: 1 Yield: 100% 1.1 Reagents: Oxygen Catalysts: Cupric chloride, Poly[[1-(cyanomethyl)[5,5'-bi-1<i>H</i>-benzimidazole]-2,2'-diyl]-1,3-phenylene] (reaction products with palladium chloride) Solvents: Ethanol, Water	Polymer-supported Pd(II) Wacker-type catalysts By: Sherrington, David C.; et al Macromolecular Symposia (1994), 80(5th International Symposium on Macromolecule-Metal Complexes), 193-213.
31-502-CAS-2552579 Steps: 1 Yield: 98% 1.1 Reagents: Di-<i>O</i>-methyl-β-cyclodextrin Catalysts: Copper sulfate, Palladium sulfate (Pd SO₄), Vanadate (9-), (octa-μ-oxohexaoxohexamolybdate)hexadeca-μ-oxohexaoxo[μ₁₂-[phosphato(3-)-κO:κO:κO:κO':κO':κO':κO'':κO'':κO'':κO'':κO'']hexa-, hydrogen (1:9) Solvents: Water 1.2 Reagents: Oxygen	A very useful and efficient Wacker oxidation of higher α-olefins in the presence of per(2,6-di-<i>O</i>-methyl)-β-cyclodextrin By: Monflier, Eric; et al Tetrahedron Letters (1995), 36(3), 387-8.
Scheme 4 (3 Reactions) Steps: 1 Yield: 97-100%  Suppliers (83)  Suppliers (98)	
31-614-CAS-24718988 Steps: 1 Yield: 100% 1.1 Reagents: 1,1,1,3,5,5,5-Heptamethyltrisiloxane Catalysts: 2725020-49-3 Solvents: Ethanol; 12 h, 60 °C 1.2 Reagents: Potassium fluoride; 1 h, rt Experimental Protocols	Aluminum Hydroxide Secondary Building Units in a Metal-Organic Framework Support Earth-Abundant Metal Catalysts for Broad-Scope Organic Transformations By: Feng, Xuanyu; et al ACS Catalysis (2019), 9(4), 3327-3337.
31-502-CAS-17865363 Steps: 1 Yield: 99% 1.1 Reagents: Oxygen Catalysts: Zirconium dioxide, Palladium (nanoparticles supported on ZrO₂) Solvents: Methanol, Dimethyl sulfoxide; 1 h, 10 atm, 80 °C 1.2 Solvents: Water; 20 h, 2 atm, 80 °C Experimental Protocols	Wacker Oxidation of Terminal Alkenes Over Zr O₂-Supported Pd Nanoparticles Under Acid- and Cocatalyst-Free Conditions By: Zhang, Zhenzhong; et al ChemSusChem (2017), 10(17), 3482-3489.
31-614-CAS-43144866 Steps: 1 Yield: 97% 1.1 Reagents: <i>tert</i>-Butyl hydroperoxide Catalysts: Palladium diacetate, 1,2-Dihydro-1-(2-pyridinyl)-3<i>H</i>-indazol-3-one Solvents: Acetonitrile, Decane, 1,1,1,3,3,3-Hexafluoro-2-propanol; 6 h, 60 °C Experimental Protocols	Covalent bidentate ligand-enabled regioselective Wacker-type oxidation of olefins By: Chen, Liping; et al RSC Advances (2024), 14(51), 37928-37932.

Scheme 5 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (20)

Suppliers (91)

31-502-CAS-5156158

Steps: 1 Yield: 100%

Ethylene Oxidation by a Platinum(II) Hydroxo Complex. Insights into the Wacker Process1.1 **Catalysts:** Platinum(2+), bis[(1,2,5,6- η)-1,5-cyclooctadiene]di- μ -hydroxydi-, 1,1,1-trifluoromethanesulfonate (1:2)**Solvents:** Dichloromethane- d_2 ; 8 h, 27 °C

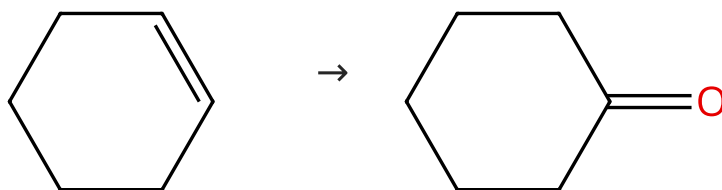
By: Weliange, Nandita M.; et al

Organometallics (2012), 31(19), 6823-6833.

Experimental Protocols

Scheme 6 (2 Reactions)

Steps: 1 Yield: 100%



Suppliers (63)

Suppliers (102)

31-502-CAS-12748837

Steps: 1 Yield: 100%

Improved method for the Wacker oxidation of cyclic and internal olefins1.1 **Reagents:** Quinone, Perchloric acid
Catalysts: Palladium diacetate
Solvents: Dimethylformamide, Water

By: Miller, D. G.; et al

Journal of Organic Chemistry (1990), 55(9), 2924-7.

31-502-CAS-15148133

Steps: 1 Yield: 100%

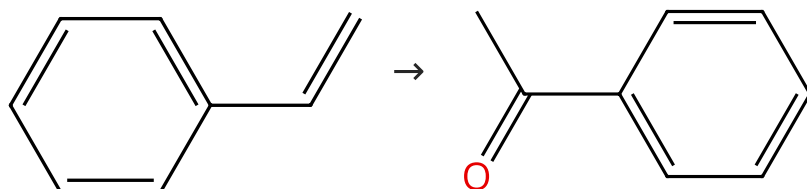
Improved method for the Wacker oxidation of cyclic and internal olefins1.1 **Reagents:** Quinone, Perchloric acid
Catalysts: Palladium diacetate
Solvents: Dimethylformamide, Water

By: Miller, D. G.; et al

Journal of Organic Chemistry (1990), 55(9), 2924-7.

Scheme 7 (8 Reactions)

Steps: 1 Yield: 96-100%



Suppliers (115)

Suppliers (105)

31-502-CAS-5979432

Steps: 1 Yield: 100%

Palladium ion-induced supramolecular gel formation of tris-urea molecules1.1 **Catalysts:** Palladium diacetate, *N,N'*,*N'''*-(2,4,6-Triethyl-1,3,5-benzenetriyl)tris(methyleneoxy-3,1-phenylene)]tris[*N*-4-pyridinylurea]**Solvents:** Dimethyl sulfoxide, Water; > 1 min, rt

By: Aoyama, Ryohei; et al

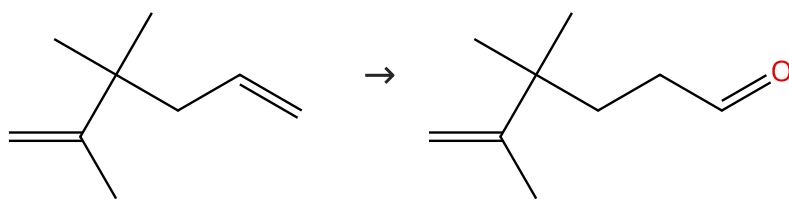
Polymer Journal (Tokyo, Japan) (2015), 47(2), 136-140.

Experimental Protocols

31-502-CAS-17861974 Steps: 1 Yield: 99% 1.1 Reagents: Oxygen Catalysts: Zirconium dioxide, Palladium (nanoparticles supported on ZrO₂) Solvents: Methanol, Dimethyl sulfoxide; 1 h, 10 atm, 80 °C 1.2 Solvents: Water; 20 h, 2 atm, 80 °C Experimental Protocols	Wacker Oxidation of Terminal Alkenes Over Zr O₂-Supported Pd Nanoparticles Under Acid- and Cocatalyst-Free Conditions By: Zhang, Zhenzhong; et al ChemSusChem (2017), 10(17), 3482-3489.
31-502-CAS-10803546 Steps: 1 Yield: 99% 1.1 Reagents: Hydrogen peroxide Catalysts: Carbamic acid, <i>N</i>-[3-(triethoxysilyl)propyl]-, <i>C,C'</i>-[(1,3-dimethyl-1,3-propanediylidene)bis(nitrilo-4,1-phenylene)] ester, polymer with silicic acid (H₄SiO₄) tetraethyl ester (complexes with palladium) Solvents: Acetonitrile, Water; 3 h, rt; 5 h, 60 °C	β-Diketimine appended periodic mesoporous organosilica as a scaffold for immobilization of palladium acetate: An efficient green catalyst for Wacker type reaction By: Borah, Parijat; et al Journal of Catalysis (2014), 318, 43-52.
31-614-CAS-35942712 Steps: 1 Yield: 98% 1.1 Reagents: <i>tert</i>-Butyl hydroperoxide Catalysts: Palladium diacetate, 2-(1<i>H</i>-Indazol-1-yl)quinoline Solvents: Decane, 1,1,1,3,3,3-Hexafluoro-2-propanol; 12 h, 60 °C Experimental Protocols	Pd-Catalyzed TBHP-Mediated Selective Wacker-Type Oxidation and Oxo-acyloxylation of Olefins Using a 2-(1<i>H</i>-Indazol-1-yl)quinoline Ligand By: Zhang, Shuaizhong; et al Organic Letters (2023), 25(11), 1850-1855.
31-614-CAS-43144893 Steps: 1 Yield: 97% 1.1 Reagents: <i>tert</i>-Butyl hydroperoxide Catalysts: Palladium diacetate, 3066476-05-6 Solvents: Acetonitrile, Decane, 1,1,1,3,3,3-Hexafluoro-2-propanol; 6 h, 60 °C Experimental Protocols	Covalent bidentate ligand-enabled regioselective Wacker-type oxidation of olefins By: Chen, Liping; et al RSC Advances (2024), 14(51), 37928-37932.
31-614-CAS-35942745 Steps: 1 Yield: 97% 1.1 Reagents: <i>tert</i>-Butyl hydroperoxide Catalysts: 2919351-23-6 Solvents: Decane, 1,1,1,3,3,3-Hexafluoro-2-propanol; 12 h, 60 °C Experimental Protocols	Pd-Catalyzed TBHP-Mediated Selective Wacker-Type Oxidation and Oxo-acyloxylation of Olefins Using a 2-(1<i>H</i>-Indazol-1-yl)quinoline Ligand By: Zhang, Shuaizhong; et al Organic Letters (2023), 25(11), 1850-1855.
31-614-CAS-43144890 Steps: 1 Yield: 96% 1.1 Reagents: <i>tert</i>-Butyl hydroperoxide Catalysts: Palladium diacetate, 3066476-04-5 Solvents: Acetonitrile, Decane, 1,1,1,3,3,3-Hexafluoro-2-propanol; 6 h, 60 °C Experimental Protocols	Covalent bidentate ligand-enabled regioselective Wacker-type oxidation of olefins By: Chen, Liping; et al RSC Advances (2024), 14(51), 37928-37932.
31-502-CAS-16689339 Steps: 1 Yield: 96% 1.1 Reagents: Oxygen Catalysts: Palladium, Cuprous chloride Solvents: Ethanol, Water; 1 atm, 75 °C Experimental Protocols	Highly selective Wacker oxidation of terminal olefins using magnetically recyclable Pd-Fe₃O₄ heterodimer nanocrystals By: Byun, Sangmoon; et al RSC Advances (2013), 3(37), 16296-16299.

Scheme 8 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (2)

Supplier (1)

31-502-CAS-11710115

Steps: 1 Yield: 99%

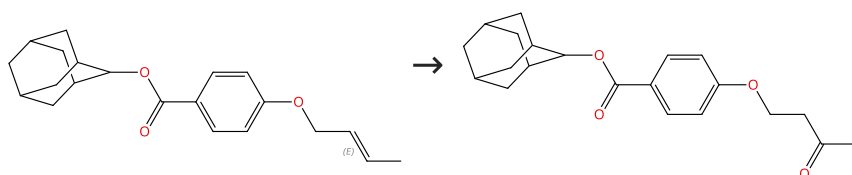
Abnormal and regioselective Wacker oxidation of 1,5-dienes

1.1 **Reagents:** Cuprous chloride, Oxygen
Catalysts: Palladium chloride
Solvents: Dimethylformamide, Water; 24 h, rt

By: Ho, Tse-Lok; et al
 Tetrahedron Letters (2003), 44(36), 6955-6957.

Scheme 9 (1 Reaction)

Steps: 1 Yield: 99%



Double bond geometry shown

Supplier (1)

31-502-CAS-21642402

Steps: 1 Yield: 99%

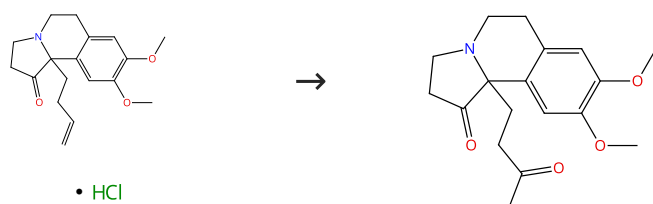
Regioselective Wacker-Type Oxidation of Internal Olefins in *t*BuOH Using Oxygen as the Sole Oxidant and *t*BuONO as the Organic Redox Cocatalyst

1.1 **Reagents:** Oxygen
Catalysts: *tert*-Butyl nitrite, Bis(benzonitrile)dichloropalladium
Solvents: *tert*-Butanol; 24 h, 25 °C
 1.2 **Solvents:** Water; 25 °C

By: Huang, Qing; et al
 Organic Letters (2020), 22(3), 965-969.

Scheme 10 (1 Reaction)

Steps: 1 Yield: 99%



• HCl

31-502-CAS-19615694

Steps: 1 Yield: 99%

Development of an Efficient Synthesis of rac-3-Demethoxyerythratidinone via a Titanium(III) Catalyzed Imine-Nitrile Coupling

1.1 **Reagents:** Quinone
Catalysts: Palladium, bis(acetonitrile)dichloro-
Solvents: Acetonitrile, Water; 24 h, 23 °C → 70 °C; 70 °C → 23 °C

By: Luu, Hieu-Trinh; et al

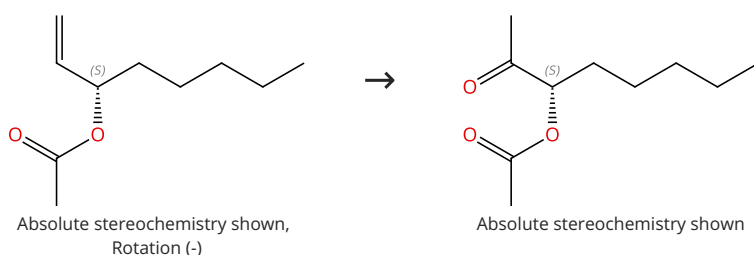
1.2 **Reagents:** Hydrochloric acid
Solvents: Diethyl ether, Water; cooled

European Journal of Organic Chemistry (2019), 2019(1), 139-149.

Experimental Protocols

Scheme 11 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (5)

31-502-CAS-579844

Steps: 1 Yield: 99%

A General and Efficient Catalyst System for a Wacker- Type Oxidation Using TBHP as the Terminal Oxidant: Application to Classically Challenging Substrates

By: Michel, Brian W.; et al

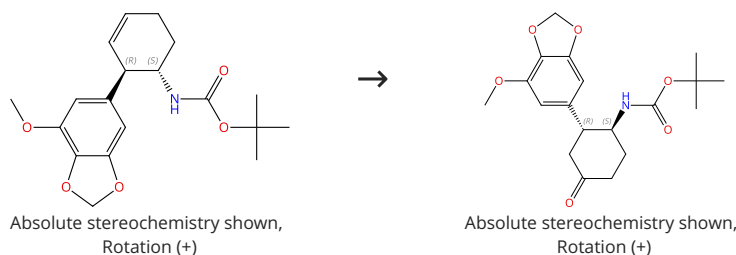
Journal of the American Chemical Society (2009), 131(17), 6076-6077.

- 1.1 **Catalysts:** Silver hexafluoroantimonate, (*SP*-4-3)-Dichloro[2-(4, 5-dihydro-2-oxazolyl- κ^N)quinoline- κ^M]palladium
Solvents: Dichloromethane; 15 min, rt
- 1.2 **Reagents:** *tert*-Butyl hydroperoxide
Solvents: Dichloromethane, Water; 10 min, rt
- 1.3 rt
- 1.4 **Reagents:** Sodium sulfite
Solvents: Water; rt

Experimental Protocols

Scheme 12 (1 Reaction)

Steps: 1 Yield: 99%



31-502-CAS-19707460

Steps: 1 Yield: 99%

Total synthesis of (+)-trans- dihydronarciclasine from (+)-7-azabicyclo[2.2.1]heptanone

By: Pandey, Ganesh; et al

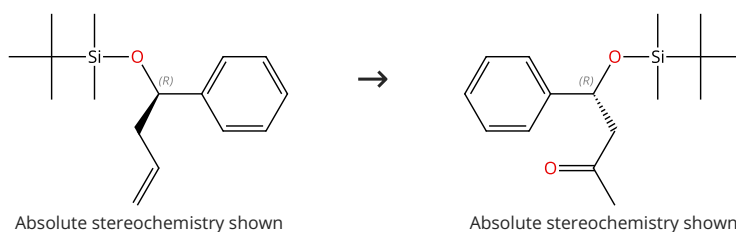
Tetrahedron (2018), 74(39), 5752-5757.

- 1.1 **Reagents:** Quinone, Sulfuric acid
Catalysts: Palladium diacetate
Solvents: Acetonitrile, Water; 24 h, rt

Experimental Protocols

Scheme 13 (1 Reaction)

Steps: 1 Yield: 99%



Scheme 14 (1 Reaction)

Steps: 1 Yield: 99%

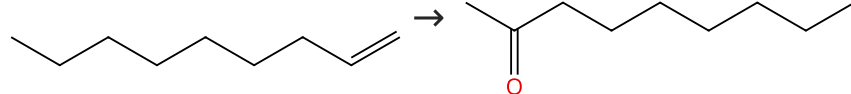
The reaction scheme shows 1-hexene (a six-carbon chain with a terminal double bond) reacting to form 2-hexanone (a six-carbon chain with a ketone group at the second carbon). The oxygen atom in the ketone group is highlighted in red.

Suppliers (68)

Suppliers (68)

31-502-CAS-4444471	Steps: 1 Yield: 99%	Phenanthroline Decorated by a Crown Ether as a Module for Metallorganic-Polyoxometalate Hybrid Catalysts: The Wacker Type Oxidation of Alkenes with Nitrous Oxide as Terminal Oxidant
1.1 Reagents: Water, Nitrous oxide Catalysts: Palladium, dichloro(2,3,5,6,8,9,11,12-octahydro-1,4,7,10,13-pentaioxacyclopentadecino[2,3-f][1,10]phenanthroline-κN¹⁷,κN¹⁸)-, (<i>SP</i>-4-2)-, compd. with hydrogen (heptadeca-μ-oxodecaoxodecamolybdate)hepta-μ-oxidioxo [μ₁₂-[phosphato(3-)-κO:κO:κO:κO:κO':κO':κO'':κO''':κO'']divanadate(5-), hydrate (1:5:1:1) Solvents: Dimethylacetamide, Water; 18 h, 3 bar, 150 °C	By: Ettedgui, Jessica; et al	Journal of the American Chemical Society (2009), 131(1), 4-5.
Experimental Protocols		

Scheme 15 (3 Reactions) Steps: 1 Yield: 97-99%



 Suppliers (59)

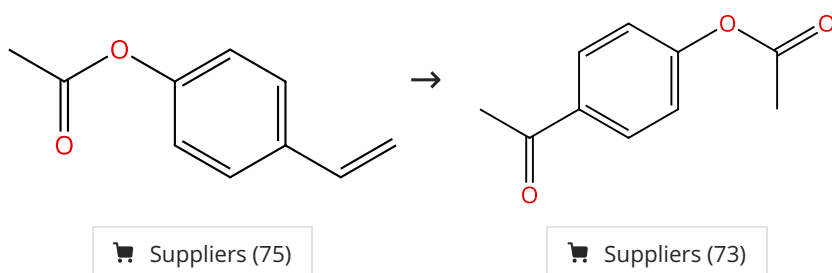
 Suppliers (77)

31-502-CAS-447110	Steps: 1 Yield: 99%	Phenanthroline Decorated by a Crown Ether as a Module for Metallorganic-Polyoxometalate Hybrid Catalysts: The Wacker Type Oxidation of Alkenes with Nitrous Oxide as Terminal Oxidant
1.1 Reagents: Water, Nitrous oxide Catalysts: Palladium, dichloro(2,3,5,6,8,9,11,12-octahydro-1,4,7,10,13-pentaoxacyclopentadecino[2,3-<i>f</i>][1,10]phenanthroline-κ^{N17},κ^{M18})-, (<i>SP</i>-4-2)-, compd. with hydrogen (heptadeca-μ-oxodecaoxodecamolybdate)hepta-μ-oxodioxo [μ₁₂-[phosphato(3-)-κ^O:κ^O:κ^O:κ^O:κ^O:κ^O:κ^O:κ^O:κ^O:κ^O:κ^O:κ^O]-divanadate(5-), hydrate (1:5:1:1) Solvents: Dimethylacetamide, Water; 18 h, 3 bar, 150 °C	By: Ettegui, Jessica; et al	Journal of the American Chemical Society (2009), 131(1), 4-5.
Experimental Protocols		

31-502-CAS-4541234 Steps: 1 Yield: 97% 1.1 Reagents: Oxygen Catalysts: Palladium (polymer- or cyclodextrin-supported)	Surface active macromolecular and supramolecular complexes: design and catalysis By: Karakhanov, Edward; et al Macromolecular Symposia (2000), 156, 137-145.
31-502-CAS-374070 Steps: 1 Yield: 97% 1.1 Reagents: Di-<i>O</i>-methyl-β-cyclodextrin Catalysts: Copper sulfate, Palladium sulfate (Pd SO₄), Vanadate (9-), (octa-μ-oxohexaoxohexamolybdate)hexadeca-μ-oxohexaoxo[μ₁₂-[phosphato(3-)-$\kappa O:\kappa O:\kappa O:\kappa O:\kappa O:\kappa O:\kappa O:\kappa O:\kappa O:\kappa O:\kappa O:\kappa O$]]hexa-, hydrogen (1:9) Solvents: Water 1.2 Reagents: Oxygen	A very useful and efficient Wacker oxidation of higher α-olefins in the presence of per(2,6-di-<i>O</i>-methyl)-β-cyclodextrin By: Monflier, Eric; et al Tetrahedron Letters (1995), 36(3), 387-8.

Scheme 16 (1 Reaction)

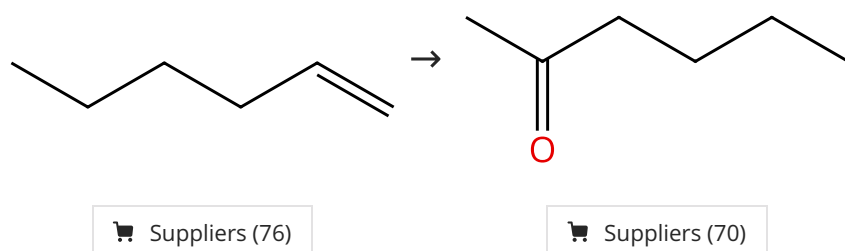
Steps: 1 Yield: 99%



31-614-CAS-35942713 Steps: 1 Yield: 99% 1.1 Reagents: <i>tert</i>-Butyl hydroperoxide Catalysts: Palladium diacetate, 2-(1<i>H</i>-Indazol-1-yl)quinoline Solvents: Decane, 1,1,1,3,3,3-Hexafluoro-2-propanol; 12 h, 60 °C	Pd-Catalyzed TBHP-Mediated Selective Wacker-Type Oxidation and Oxo-acyloxylation of Olefins Using a 2-(1<i>H</i>-Indazol-1-yl)quinoline Ligand By: Zhang, Shuaizhong; et al Organic Letters (2023), 25(11), 1850-1855.
Experimental Protocols	

Scheme 17 (2 Reactions)

Steps: 1 Yield: 99%

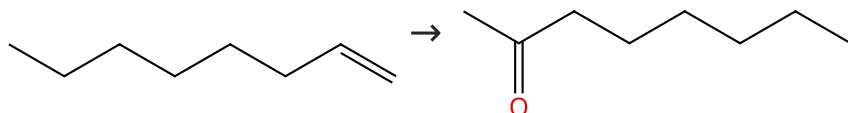


31-502-CAS-598348 Steps: 1 Yield: 99% 1.1 Reagents: Water, Nitrous oxide Catalysts: Palladium, dichloro(2,3,5,6,8,9,11,12-octahydro-1,4,7,10,13-pentaoxacyclopentadecino[2,3-<i>f</i>][1,10]phenanthroline-$\kappa N^{17}, \kappa N^{18}$)-, (<i>SP</i>-4-2)-, compd. with hydrogen (heptadeca-μ-oxodecaoxodecamolybdate)hepta-μ-oxodioxo[μ₁₂-[phosphato(3-)-$\kappa O:\kappa O:\kappa O:\kappa O:\kappa O:\kappa O:\kappa O:\kappa O:\kappa O:\kappa O:\kappa O:\kappa O$]]divanadate(5-), hydrate (1:5:1:1) Solvents: Dimethylacetamide, Water; 18 h, 3 bar, 150 °C	Phenanthroline Decorated by a Crown Ether as a Module for Metallorganic-Polyoxometalate Hybrid Catalysts: The Wacker Type Oxidation of Alkenes with Nitrous Oxide as Terminal Oxidant By: Ettegui, Jessica; et al Journal of the American Chemical Society (2009), 131(1), 4-5.
Experimental Protocols	

31-502-CAS-4909647	Steps: 1 Yield: 99%	Supercritical carbon dioxide and poly(ethylene glycol): An environmentally benign biphasic solvent system for aerobic oxidation of styrene
1.1 Reagents: Oxygen Catalysts: Palladium chloride, Cuprous chloride Solvents: Carbon dioxide, Polyethylene glycol; rt → 60 °C; 24 h, 16 MPa, 60 °C; cooled		By: Wang, Jin-Quan; et al Green Chemistry (2007), 9(8), 882-887.
Experimental Protocols		

Scheme 18 (2 Reactions)

Steps: 1 Yield: 98-99%



Suppliers (93)

Suppliers (87)

31-502-CAS-13865385	Steps: 1 Yield: 99%	Phenanthroline Decorated by a Crown Ether as a Module for Metallorganic-Polyoxometalate Hybrid Catalysts: The Wacker Type Oxidation of Alkenes with Nitrous Oxide as Terminal Oxidant
1.1 Reagents: Water, Nitrous oxide Catalysts: Palladium, dichloro(2,3,5,6,8,9,11,12-octahydro-1,4,7,10,13-pentaoxacyclopentadecino[2,3- <i>f</i>][1,10]phenanthroline- $\kappa N^{17}, \kappa N^{18}$)-, (<i>SP</i> -4-2)-, compd. with hydrogen (heptadeca- μ -oxodecaoxodecamolybdate)hepta- μ -oxodioxo [μ_{12} -[phosphato(3-)- $\kappa O: \kappa O: \kappa O: \kappa O: \kappa O: \kappa O: \kappa O: \kappa O: \kappa O: \kappa O: \kappa O: \kappa O$]]divanadate(5-), hydrate (1:5:1:1) Solvents: Dimethylacetamide, Water; 18 h, 3 bar, 150 °C		By: Ettedgui, Jessica; et al Journal of the American Chemical Society (2009), 131(1), 4-5.
Experimental Protocols		
31-502-CAS-6542687	Steps: 1 Yield: 98%	A very useful and efficient Wacker oxidation of higher α -olefins in the presence of per(2,6-di-O-methyl)- β -cyclodextrin
1.1 Reagents: Di- <i>O</i> -methyl- β -cyclodextrin Catalysts: Copper sulfate, Palladium sulfate (Pd SO ₄), Vanadate (9-), (octa- μ -oxohexaoxohexamolybdate)hexadeca- μ -oxohexaoxo[μ_{12} -[phosphato(3-)- $\kappa O: \kappa O: \kappa O: \kappa O: \kappa O: \kappa O: \kappa O: \kappa O: \kappa O: \kappa O: \kappa O: \kappa O$]]hexa-, hydrogen (1:9) Solvents: Water		By: Monflier, Eric; et al Tetrahedron Letters (1995), 36(3), 387-8.
1.2 Reagents: Oxygen		

Scheme 19 (1 Reaction)

Steps: 1 Yield: 98%



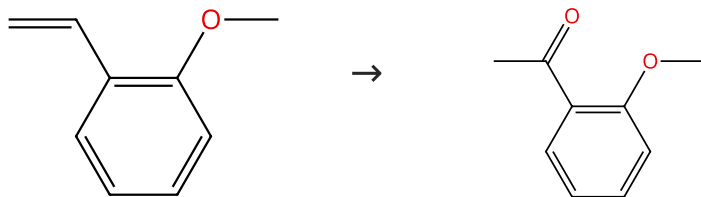
Suppliers (66)

Suppliers (81)

31-502-CAS-16691776	Steps: 1 Yield: 98%	Highly selective Wacker oxidation of terminal olefins using magnetically recyclable Pd-Fe ₃ O ₄ heterodimer nanocrystals
1.1 Reagents: Oxygen Catalysts: Palladium, Cuprous chloride Solvents: Ethanol, Water; 1 atm, 75 °C		By: Byun, Sangmoon; et al RSC Advances (2013), 3(37), 16296-16299.
Experimental Protocols		

Scheme 20 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (53)

Suppliers (87)

31-502-CAS-17865365

Steps: 1 Yield: 98%

Wacker Oxidation of Terminal Alkenes Over Zr O₂-Supported Pd Nanoparticles Under Acid- and Cocatalyst-Free Conditions

By: Zhang, Zhenzhong; et al

ChemSusChem (2017), 10(17), 3482-3489.

1.1 Reagents: Oxygen

Catalysts: Zirconium dioxide, Palladium (nanoparticles supported on ZrO₂)

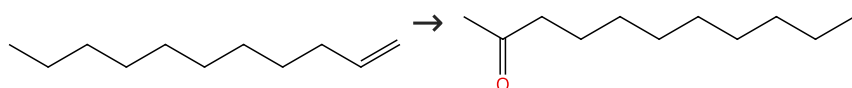
Solvents: Methanol, Dimethyl sulfoxide; 1 h, 10 atm, 80 °C

1.2 Solvents: Water; 20 h, 2 atm, 80 °C

Experimental Protocols

Scheme 21 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (87)

Suppliers (87)

31-502-CAS-4691065

Steps: 1 Yield: 98%

A very useful and efficient Wacker oxidation of higher α -olefins in the presence of per(2,6-di-O-methyl)- β -cyclodextrin

By: Monflier, Eric; et al

Tetrahedron Letters (1995), 36(3), 387-8.

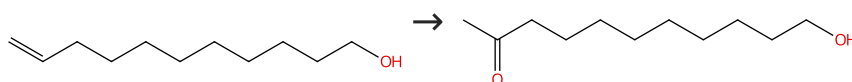
1.1 Reagents: Di-O-methyl- β -cyclodextrinCatalysts: Copper sulfate, Palladium sulfate (Pd SO₄), Vanadate (9-), (octa- μ -oxohexaoxohexamolybdate)hexadeca- μ -oxohexaoxo[μ ₁₂-[phosphato(3-)- κ O: κ O: κ O: κ O': κ O': κ O'': κ O'': κ O'': κ O'']]hexa-, hydrogen (1:9)

Solvents: Water

1.2 Reagents: Oxygen

Scheme 22 (1 Reaction)

Steps: 1 Yield: 98%



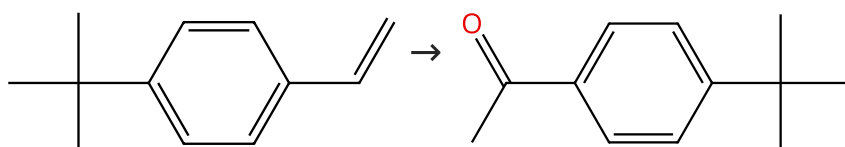
Suppliers (88)

Suppliers (12)

31-502-CAS-2371044	Steps: 1 Yield: 98%	A General and Efficient Catalyst System for a Wacker- Type Oxidation Using TBHP as the Terminal Oxidant: Application to Classically Challenging Substrates
1.1 Catalysts: Silver hexafluoroantimonate, (<i>SP</i> -4-3)-Dichloro[2-(4, 5-dihydro-2-oxazolyl-κ ^N 3)quinoline-κ ^N]palladium Solvents: Dichloromethane; 15 min, rt		By: Michel, Brian W.; et al
1.2 Reagents: <i>tert</i> -Butyl hydroperoxide Solvents: Dichloromethane, Water; 10 min, rt; rt → 0 °C		Journal of the American Chemical Society (2009), 131(17), 6076-6077.
1.3 5 min, 0 °C; 35 min, 0 °C → rt		
1.4 Reagents: Sodium sulfite Solvents: Water; rt		
Experimental Protocols		

Scheme 23 (1 Reaction)

Steps: 1 Yield: 98%



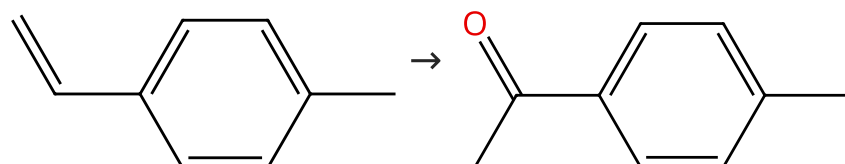
Suppliers (59)

Suppliers (95)

31-614-CAS-35942722	Steps: 1 Yield: 98%	Pd-Catalyzed TBHP-Mediated Selective Wacker-Type Oxidation and Oxo-acyloxylation of Olefins Using a 2- (1<i>H</i>-Indazol-1-yl)quinoline Ligand
1.1 Reagents: <i>tert</i> -Butyl hydroperoxide Catalysts: Palladium diacetate, 2-(1 <i>H</i> -Indazol-1-yl)quinoline Solvents: Decane, 1,1,1,3,3,3-Hexafluoro-2-propanol; 12 h, 60 °C		By: Zhang, Shuaizhong; et al
Experimental Protocols		Organic Letters (2023), 25(11), 1850-1855.

Scheme 24 (3 Reactions)

Steps: 1 Yield: 97-98%



Suppliers (66)

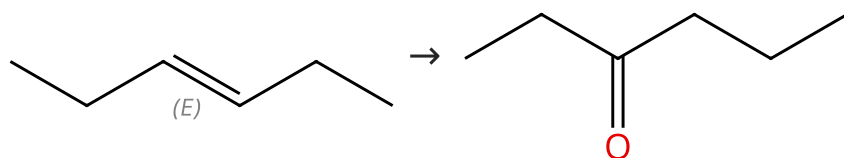
Suppliers (103)

31-614-CAS-43144862	Steps: 1 Yield: 98%	Covalent bidentate ligand-enabled regioselective Wacker-type oxidation of olefins
1.1 Reagents: <i>tert</i> -Butyl hydroperoxide Catalysts: Palladium diacetate, 1,2-Dihydro-1-(2-pyridinyl)-3 <i>H</i> -indazol-3-one Solvents: Acetonitrile, Decane, 1,1,1,3,3,3-Hexafluoro-2-propanol; 6 h, 60 °C		By: Chen, Liping; et al
Experimental Protocols		RSC Advances (2024), 14(51), 37928-37932.
31-502-CAS-17865359	Steps: 1 Yield: 98%	Wacker Oxidation of Terminal Alkenes Over Zr O₂-Supported Pd Nanoparticles Under Acid- and Cocatalyst-Free Conditions
1.1 Reagents: Oxygen Catalysts: Zirconium dioxide, Palladium (nanoparticles supported on ZrO ₂) Solvents: Methanol, Dimethyl sulfoxide; 1 h, 10 atm, 80 °C		By: Zhang, Zhenzhong; et al
1.2 Solvents: Water; 20 h, 2 atm, 80 °C		ChemSusChem (2017), 10(17), 3482-3489.
Experimental Protocols		

31-502-CAS-19652298	Steps: 1 Yield: 97%	Efficient Palladium(0) supported on reduced graphene oxide for selective oxidation of olefins using graphene oxide as a 'solid weak acid'
1.1 Catalysts: Palladium, Graphene (graphene oxide) Solvents: Acetonitrile, Water; 30 min, 25 °C		By: Gao, Xi; et al
1.2 Reagents: Hydrogen peroxide; 12 h, 55 °C		Catalysis Communications (2019), 122, 73-78.

Scheme 25 (1 Reaction)

Steps: 1 Yield: 98%



Double bond geometry shown

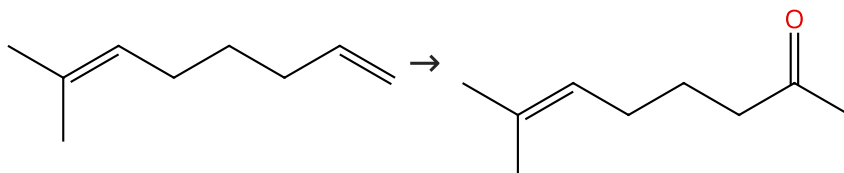
Suppliers (51)

Suppliers (48)

31-502-CAS-12209272	Steps: 1 Yield: 98%	Wacker-Type Oxidation of Internal Olefins Using a Pd Cl ₂ /N,N-Dimethylacetamide Catalyst System under Copper-Free Reaction Conditions
1.1 Reagents: Oxygen Catalysts: Palladium chloride Solvents: Dimethylacetamide, Water; 1 h, 9 atm, rt		By: Mitsudome, Takato; et al
1.2 Reagents: Oxygen; 10 h, 3 atm, 80 °C		Angewandte Chemie, International Edition (2010), 49(7), 1238-1240, S1238/1-S1238/6.
Experimental Protocols		

Scheme 26 (1 Reaction)

Steps: 1 Yield: 97%



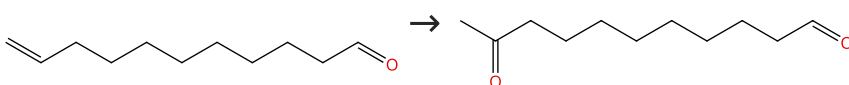
Suppliers (13)

Suppliers (4)

31-502-CAS-3358113	Steps: 1 Yield: 97%	Selective Wacker-type oxidation of terminal alkenes and dienes using the Pd(II)/molybdovanadophosphate (NPMoV)/O ₂ system
1.1 Reagents: Oxygen Catalysts: Methanesulfonic acid, Hydroquinone, Palladium diacetate, Sodium chloride, Molybdovanadophosphate Solvents: Ethanol, Water; 5 h, rt		By: Yokota, Takahiro; et al
		Tetrahedron Letters (2002), 43(49), 8887-8891.

Scheme 27 (1 Reaction)

Steps: 1 Yield: 97%



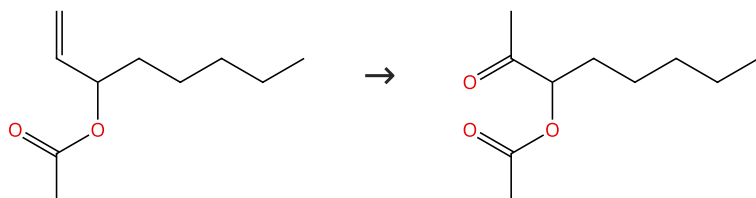
Suppliers (68)

Suppliers (2)

31-502-CAS-11984856	Steps: 1 Yield: 97%	Tsuji-Wacker Oxidation of Terminal Olefins using a Palladium-Carbon Nanotube Nanohybrid By: Donck, Simon; et al ChemCatChem (2015), 7(15), 2318-2322.
1.1 Reagents: Oxygen Catalysts: Palladium, Cuprous chloride Solvents: Tetrahydrofuran, Water; 15 min, rt		
1.2 36 h, rt		
Experimental Protocols		

Scheme 28 (1 Reaction)

Steps: 1 Yield: 97%



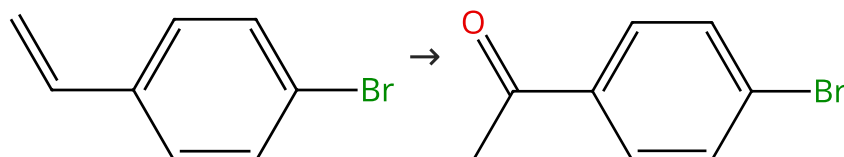
Suppliers (44)

Suppliers (2)

31-502-CAS-7378884	Steps: 1 Yield: 97%	On the Mechanism of the Palladium-Catalyzed <i>tert</i>-Butylhydroperoxide-Mediated Wacker-Type Oxidation of Alkenes Using Quinoline-2-Oxazoline Ligands By: Michel, Brian W.; et al Journal of the American Chemical Society (2011), 133(21), 8317-8325.
1.1 Reagents: <i>tert</i> -Butyl hydroperoxide, Silver hexafluoro antimonate Catalysts: Palladium, bis(acetonitrile)dichloro-, 2-(4,5-Dihydro-5,5-dimethyl-2-oxazolyl)quinoline Solvents: Dichloromethane, Water; 16 h		
Experimental Protocols		

Scheme 29 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (86)

Suppliers (84)

31-614-CAS-35942718	Steps: 1 Yield: 97%	Pd-Catalyzed TBHP-Mediated Selective Wacker-Type Oxidation and Oxo-acyloxylation of Olefins Using a 2-(1<i>H</i>-Indazol-1-yl)quinoline Ligand By: Zhang, Shuaizhong; et al Organic Letters (2023), 25(11), 1850-1855.
1.1 Reagents: <i>tert</i> -Butyl hydroperoxide Catalysts: Palladium diacetate, 2-(1 <i>H</i> -Indazol-1-yl)quinoline Solvents: Decane, 1,1,1,3,3,3-Hexafluoro-2-propanol; 12 h, 60 °C		
Experimental Protocols		

Scheme 30 (1 Reaction)

Steps: 1 Yield: 97%



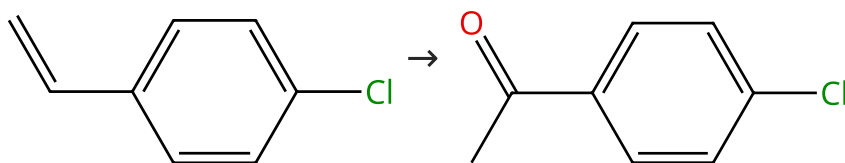
Suppliers (59)

Suppliers (86)

31-502-CAS-16691774	Steps: 1 Yield: 97%	Highly selective Wacker oxidation of terminal olefins using magnetically recyclable Pd-Fe ₃ O ₄ heterodimer nanocrystals
1.1 Reagents: Oxygen Catalysts: Palladium, Cuprous chloride Solvents: Ethanol, Water; 1 atm, 75 °C		By: Byun, Sangmoon; et al RSC Advances (2013), 3(37), 16296-16299.
Experimental Protocols		

Scheme 31 (1 Reaction)

Steps: 1 Yield: 97%



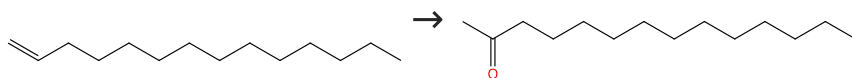
Suppliers (78)

Suppliers (67)

31-614-CAS-35942710	Steps: 1 Yield: 97%	Pd-Catalyzed TBHP-Mediated Selective Wacker-Type Oxidation and Oxo-acyloxylation of Olefins Using a 2-(1H-Indazol-1-yl)quinoline Ligand
1.1 Reagents: <i>tert</i> -Butyl hydroperoxide Catalysts: Palladium diacetate, 2-(1 <i>H</i> -Indazol-1-yl)quinoline Solvents: Decane, 1,1,1,3,3,3-Hexafluoro-2-propanol; 12 h, 60 °C		By: Zhang, Shuaizhong; et al Organic Letters (2023), 25(11), 1850-1855.
Experimental Protocols		

Scheme 32 (1 Reaction)

Steps: 1 Yield: 96%



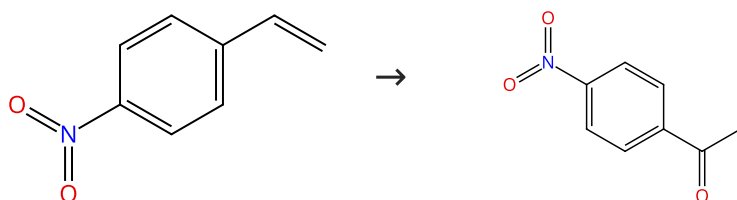
Suppliers (71)

Suppliers (63)

31-502-CAS-14609573	Steps: 1 Yield: 96%	Synthesis of methyl ketones from terminal olefins using Pd Cl ₂ /CrO ₃ system mimicking the Wacker process
1.1 Reagents: Chromium trioxide Catalysts: Palladium chloride Solvents: Acetonitrile, Water; rt → 60 °C; 6.5 h, 60 °C		By: Fernandes, Rodney A.; et al Tetrahedron (2014), 70(32), 4760-4767.

Scheme 33 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (64)

Suppliers (79)

31-614-CAS-38008194	Steps: 1 Yield: 96%	Catalyst-Controlled Selectivity in Oxidation of Olefins: Highly Facile Success to Functionalized Aldehydes and Ketones
1.1 Reagents: Hydrogen peroxide, Oxygen Catalysts: 2760337-75-3 Solvents: Water; 80 °C		By: Zheng, Kai; et al Catalysis Letters (2022), 152(11), 3332-3337.
Experimental Protocols		

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